MILITARY SPECIFICATION

PROPELLANT, M9
(FOR USE IN M19A2 SIGNAL)

1. SCOPE

1.1 This specification covers the M9 propellant for use in M19A2 Signal.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-652 — Propellants, Solid for Cannons Requirements and Packing.

MIL-STD-1168 — Lot Numbering of Ammunition.

DRAWING

ARMY

78-0-37 — Signal, Illumination, Ground, Star, Parachute (For use with Grenade Launcher, M7 and M8) Assembly.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Publications.** The following document forms a part of this specification to the extent herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and Revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publications should cite 49 CFR 71-90 (latest revision).

3. REQUIREMENTS

3.1 **Material.** The propellant shall be M9, flake and conforming to the requirements of MIL-STD-652.

3.2 **Ballistics.** The propellant, when fired in the following weapon and components, shall comply with the applicable requirements when tested as specified in 4.3.1:

Caliber .30 US Rifle, Model 1903
Case, Cartridge, Caliber .30, M1
Bullet, Ball, Caliber .30, M2

FSC 1375

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3.2.1 *Velocity and pressure.* At 70 degrees Fahrenheit (°F), a weight of propellant as prescribed herein shall impart a muzzle velocity to the round and produce a rate pressure as specified in table I (see 4.2.3.1).

TABLE I

No. of Projectiles	Muzzle Velocity feet per second (FPS)		Pressure pounds per square inch (psi)	
30	Minimum (min) 2240	Maximum (max) 2345	Min 37,800	Max 45,100

3.2.2 *Velocity.* The standard deviation in velocity for the lot at 70°F. shall not exceed 17.6 feet per second.

3.2.3 *Uniformity of maximum pressure.* The average maximum pressure produced, plus 2.65 times the standard deviation of the maximum pressure obtained in the uniformity series, shall not be greater than the requirements specified in table I.

3.3 **Workmanship.** The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL–STD–109 in order to define terms used herein.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of batches of product inspected.

(c) Results obtained, by defect code, for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is

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controlled by Government specifications referenced in any of the contractual documents.

(f) Number of pounds in the lot.

(g) Date submitted.

(h) Eight copies of propellant description sheet, AMC Form 1047, (00 Form 1204), containing the applicable data, (see 6.5).

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of one or more batches of M9 propellant produced by one manufacturer, in accordance with the same specification and same specification revision under one continuous set of operating conditions. Each lot shall consist of that quantity of M9 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. The criteria and procedures for the assignment of inspection lot interfix numbers shall be in accordance with MIL–STD–1168.

4.2.2 *Testing.*

4.2.2.1 *Ballistics* (see 3.2). Sufficient propellant to load thirty (30) sample rounds for rifle test shall be randomly selected from the lot for charge establishment, determining velocity and determining uniformity of maximum pressure.

4.2.2.1.1 *Pressure (see 3.2.1) Critical defect, Code No. 01001 (see 6.4).* If the mean of the maximum pressures of the test rounds exceeds the applicable requirement, the lot shall be rejected.

4.2.2.1.2 *Velocity (see 3.2.1 and 3.2.2) Major defect, Code No. 02001.* If the product of the factor 0.82 times the velocity standard deviation of the test rounds exceeds the applicable requirement, the lot shall be rejected (see 6.2).

4.2.2.1.3 *Uniformity of maximum pressure (see 3.2.3) Critical defect, Code No. 03001.* Using the rifle components specified (see 3.2) with the established propellant charge, the sample rounds shall be fired for pressure requirements utilizing the standard Frankford Arsenal pressure gage and Frankford Arsenal coppers of zero initial compression. The lot of propellant shall be rejected if it fails to meet the applicable requirements (see 6.2).

4.2.2.2 *Retest.* If for any reasons the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional charges as required.

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4.3 Test methods and procedures.

4.3.1 *Ballistic test.* The velocity barrel or pressure gage shall be set up in the mechanical stand; washed with water, and wiped dry before using for test. Three warming shots shall be fired at intervals of one minute, using small charges of the propellant to be tested. The firing of the test cartridges shall begin not more than 5 minutes after firing the last warming shot. In each of the required velocity and pressure tests, consecutive shots shall be fired at intervals of not more than 2 minutes. All tests shall be made with propellant at a temperature of 70° ± 2°F. The weight of charge loaded in each sample round for the applicable type of propellant under test shall be as follows:

24.0 ± 0.5 grains

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. The propellant shall be packed and marked in accordance with MIL–STD–652.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Packing required (see 5.1).

(c) *Process.* Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Standard deviation. The standard deviation shall be calculated with (N–1) as the divisor in a standard statistical technique equivalent to that shown in MIL–STD–414, Section B, Example B–1.

6.3 Submission of test data. In addition to the normal distribution of records, when the propellant is procured by the Department of the ARMY, one copy of all ballistic test data, Form AMC 1060, will be forwarded to the following: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–DC8, Dover, N. J.

6.4 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.5 Submission of AMC Form 1047. Information contained in propellant description sheet, AMC Form 1047, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agents is adhered to (see 4.1.1).

Custodian:

Army—MU

Preparing activity:

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Project No. 1375–A964

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